

# Disproof of Conjecture 00000000438: the degree-3 primitives of **Sym** form a 2-dimensional space, but $A(3, 1) \in \{1, 4\}$

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## Abstract

Conjecture 00000000438 asserts that the primitive Lie algebra of the Hopf algebra **Sym** of noncommutative symmetric functions has  $n$ -th homogeneous dimension equal to the Eulerian number  $A(n, 1)$ . We show that the degree-3 primitive space  $\text{Prim}(\mathbf{Sym})_3$  has dimension exactly 2, with basis  $\Psi_3$  and  $[S_2, S_1]$ . The Eulerian number  $A(3, 1)$ , however, is 4 in the convention “permutations with one descent” and 1 in the 1-based convention of OEIS A008292. In fact 2 is not of the form  $A(m, 1)$  for any  $m$  in either convention, so no shift of the index can save the formula. A self-contained Lean 4 development (core library only) builds **Sym** in degree 3 from its defining coproduct, proves that the primitive space has dimension 2, defines the Eulerian numbers by counting descents and ascents of permutations, and proves that the dimension clause is false. More generally the dimensions are 1, 1, 2, 3, 6, 9, 18, 30, 56, 99 for  $n = 1, \dots, 10$ , which match  $A(n, 1)$  only at  $n = 2$  (both conventions) and at  $n = 1$  (1-based convention).

## 1 The conjecture and the clause we refute

The statement reads:

*Definition:* The copying structure of noncommutative symmetric functions **Sym**. *Conjecture:* As a coalgebra, the primitive Lie algebra of **Sym** is isomorphic to the generalized Lie  $n$ -algebra, whose  $n$ -th homogeneous dimension equals the Eulerian number  $A(n, 1)$  (verifiable for  $n \leq 10$ ).

The Chinese version says the same. The “copying structure” is the coproduct (comultiplication) of **Sym**, and its “primitive Lie algebra” is  $\text{Prim}(\mathbf{Sym}) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}$ , a graded Lie algebra under the commutator. The phrase “generalized Lie  $n$ -algebra” names no standard object and is not defined in the statement. The only checkable content is the dimension formula, and it is meant to be checked on the primitive Lie algebra itself (“verifiable for  $n \leq 10$ ”). An isomorphism of graded objects preserves homogeneous dimensions, so the conjecture implies

$$\dim \text{Prim}(\mathbf{Sym})_n = A(n, 1) \quad \text{for all } n \geq 1. \tag{1}$$

We refute (1). If the conjecture is read as a conjunction (“ $\text{Prim}(\mathbf{Sym}) \cong L$ ” and “ $\dim L_n = A(n, 1)$ ”), a graded isomorphism transfers the second conjunct to  $\text{Prim}(\mathbf{Sym})$ , so refuting (1) refutes the conjecture. The isomorphism part is therefore never needed.

## Readings covered.

- *Eulerian convention.*  $A(n, k)$  is the number of permutations of  $\{1, \dots, n\}$  with  $k$  descents (Graham–Knuth–Patashnik), giving  $A(3, 1) = 4$ . In the 1-based convention (OEIS A008292,  $T(n, k)$  = permutations with  $k$  ascending runs, i.e.  $k - 1$  descents),  $A(3, 1) = 1$ . Ascents give the same numbers as descents. Section 4 shows that  $A(m, 1) \in \{2^m - m - 1, 1\}$  is never 2. So  $\dim \text{Prim}(\mathbf{Sym})_3 = 2$  refutes (1) under every convention and every index shift  $n \mapsto n + s$ .
- *Coefficients.* We work over  $\mathbb{Q}$  (any field of characteristic 0), the standard setting. The degree-3 computation is valid over every commutative ring: there  $\text{Prim}(\mathbf{Sym})_3$  is free of rank 2 (Remark 3).
- *Which algebra.* The statement explicitly says *noncommutative* symmetric functions, i.e.  $\mathbf{Sym}$  of Gelfand, Krob, Lascoux, Leclerc, Retakh and Thibon [1]. For the commutative ring of symmetric functions, or for  $\text{QSym}$ , every primitive space is 1-dimensional (spanned by  $p_n$ , respectively  $M_{(n)}$ ). That would agree with the degenerate 1-based value  $A(n, 1) = 1$ , but it is not the algebra named in the conjecture.

## 2 The Hopf algebra $\mathbf{Sym}$

$\mathbf{Sym} = \mathbb{Q}\langle S_1, S_2, \dots \rangle$  is the free associative algebra on noncommuting generators  $S_k$  of degree  $k$ . Its degree- $n$  part  $\mathbf{Sym}_n$  has basis  $S^I = S_{i_1} S_{i_2} \cdots S_{i_\ell}$ , where  $I = (i_1, \dots, i_\ell)$  runs over the  $2^{n-1}$  compositions of  $n$ . The coproduct is the algebra morphism  $\Delta : \mathbf{Sym} \rightarrow \mathbf{Sym} \otimes \mathbf{Sym}$  with

$$\Delta S_k = \sum_{i+j=k} S_i \otimes S_j \quad (S_0 = 1),$$

so  $\Delta S^I = \Delta S_{i_1} \cdots \Delta S_{i_\ell}$  [1, §3]. The power sums of the first kind  $\Psi_k$  are defined by Newton's formula  $kS_k = \sum_{j=0}^{k-1} S_j \Psi_{k-j}$ . They are primitive, and  $\text{Prim}(\mathbf{Sym})$  is the free Lie algebra generated by  $\Psi_1, \Psi_2, \dots$  [1]. We do not need this structure theorem for degree 3, which we compute from scratch.

Write  $\bar{\Delta}x = \Delta x - x \otimes 1 - 1 \otimes x$  for  $x$  of positive degree. Then  $x$  is primitive iff  $\bar{\Delta}x = 0$ , and  $\bar{\Delta}$  is linear.

## 3 The degree-3 primitives

**Lemma 1.** *In degree 3 (basis  $S_3, S_{21}, S_{12}, S_{111}$ , where  $S_{21} = S_2 S_1$  etc.):*

$$\begin{aligned} \bar{\Delta} S_3 &= S_2 \otimes S_1 + S_1 \otimes S_2, \\ \bar{\Delta} S_{21} &= \bar{\Delta} S_{12} = S_2 \otimes S_1 + S_1 \otimes S_2 + S_{11} \otimes S_1 + S_1 \otimes S_{11}, \\ \bar{\Delta} S_{111} &= 3 S_{11} \otimes S_1 + 3 S_1 \otimes S_{11}. \end{aligned}$$

*Proof.* Expand the products. For example,  $\Delta S_{21} = (S_2 \otimes 1 + S_1 \otimes S_1 + 1 \otimes S_2)(S_1 \otimes 1 + 1 \otimes S_1) = S_{21} \otimes 1 + S_2 \otimes S_1 + S_{11} \otimes S_1 + S_1 \otimes S_{11} + S_1 \otimes S_2 + 1 \otimes S_{21}$ .  $\Delta S_{12}$  gives the same six terms, with  $S_{12}$  in place of  $S_{21}$  in the two outer ones.  $\Delta S_{111} = (S_1 \otimes 1 + 1 \otimes S_1)^3$  has middle terms  $3S_{11} \otimes S_1 + 3S_1 \otimes S_{11}$ .  $\square$

**Theorem 2.**  $x = aS_3 + bS_{21} + cS_{12} + dS_{111}$  is primitive iff

$$a + b + c = 0 \quad \text{and} \quad b + c + 3d = 0.$$

Hence  $\text{Prim}(\mathbf{Sym})_3 = \{d(3, 0, -3, 1) + b(0, 1, -1, 0)\}$ , and  $\dim_{\mathbb{Q}} \text{Prim}(\mathbf{Sym})_3 = 2$ . A basis is

$$\Psi_3 = 3S_3 - S_{21} - 2S_{12} + S_{111}, \quad [S_2, S_1] = S_{21} - S_{12}.$$

*Proof.* By Lemma 1,  $\bar{\Delta}x = (a + b + c)(S_2 \otimes S_1 + S_1 \otimes S_2) + (b + c + 3d)(S_{11} \otimes S_1 + S_1 \otimes S_{11})$ . The four tensors  $S_J \otimes S_K$  are distinct basis elements of  $\mathbf{Sym} \otimes \mathbf{Sym}$ , so  $\bar{\Delta}x = 0$  iff both brackets vanish. The two equations have rank 2, with solutions  $a = 3d$ ,  $c = -b - 3d$  for free  $b, d$ , so the kernel has dimension  $4 - 2 = 2$ .  $\Psi_3(a, b, c, d = 3, -1, -2, 1)$  and  $[S_2, S_1](0, 1, -1, 0)$  both satisfy the equations. They are independent: the  $S_3$ -coordinate forces the coefficient of  $\Psi_3$  to be 0, and then the  $S_{21}$ -coordinate forces the other one to be 0. The expression for  $\Psi_3$  follows from Newton's formula:  $\Psi_1 = S_1$ ,  $\Psi_2 = 2S_2 - S_{11}$ ,  $3S_3 = \Psi_3 + S_1\Psi_2 + S_2\Psi_1 = \Psi_3 + 2S_{12} - S_{111} + S_{21}$ .  $\square$

*Remark 3.* The proof uses no division. Over any commutative ring  $R$  the module  $\text{Prim}(\mathbf{Sym}_R)_3$  is free with basis  $\{(3, 0, -3, 1), (0, 1, -1, 0)\}$ . In particular its dimension is 2 over every field.

## 4 Eulerian numbers

**Lemma 4.** For  $m \geq 1$ , the number of permutations of  $\{1, \dots, m\}$  with exactly one descent is  $2^m - m - 1$ . Exactly one permutation (the identity) has no descent. By the reversal  $w \mapsto w_m \cdots w_1$ , the same holds for ascents.

*Proof.* A permutation with at most one descent is an increasing word on a set  $T$  followed by an increasing word on its complement. It is determined by  $T$ , giving  $2^m$  choices. It has no descent iff  $T = \{1, \dots, k\}$  for some  $0 \leq k \leq m$ , which gives  $m + 1$  choices that all produce the identity. Every other  $T$  gives a distinct permutation with exactly one descent.  $\square$

So  $A(m, 1) = 2^m - m - 1$  in the descent convention, giving 0, 1, 4, 11, 26, 57, 120, 247, 502, 1013 for  $m = 1, \dots, 10$ , and  $A(m, 1) = 1$  for all  $m \geq 1$  in the 1-based convention. Since  $2^{m+1} - (m + 1) - 1 - (2^m - m - 1) = 2^m - 1 \geq 0$ , the sequence  $2^m - m - 1$  is nondecreasing. It takes the values 0, 0, 1, 4 at  $m = 0, 1, 2, 3$ , so it never equals 2. Thus

**Corollary 5.**  $\dim \text{Prim}(\mathbf{Sym})_3 = 2 \neq A(m, 1)$  for every  $m \geq 0$  and every convention above. In particular (1) fails at  $n = 3$ :  $2 \neq 4$  and  $2 \neq 1$ .

## 5 All degrees up to 10

$\mathbf{Sym}$  is the universal enveloping algebra of the free Lie algebra  $\text{Prim}(\mathbf{Sym})$  generated by  $\Psi_1, \Psi_2, \dots$ . By Poincaré–Birkhoff–Witt,  $\prod_{k \geq 1} (1 - t^k)^{-p_k} = \sum_n \dim \mathbf{Sym}_n t^n = \frac{1-t}{1-2t}$ , where  $p_k = \dim \text{Prim}(\mathbf{Sym})_k$ . Solving for  $p_k$  gives the first row of the table below. The same values come from a direct kernel computation and from duality with  $\text{QSym}$  (Section 7).

| $n$                                | 1 | 2 | 3 | 4  | 5  | 6  | 7   | 8   | 9   | 10   |
|------------------------------------|---|---|---|----|----|----|-----|-----|-----|------|
| $\dim \text{Prim}(\mathbf{Sym})_n$ | 1 | 1 | 2 | 3  | 6  | 9  | 18  | 30  | 56  | 99   |
| $A(n, 1)$ , one descent            | 0 | 1 | 4 | 11 | 26 | 57 | 120 | 247 | 502 | 1013 |
| $A(n, 1)$ , 1-based                | 1 | 1 | 1 | 1  | 1  | 1  | 1   | 1   | 1   | 1    |

Agreement with both conventions holds only at  $n = 2$ . At  $n = 1$  only the 1-based value agrees. From  $n = 3$  on, the formula fails in every degree up to 10 under both conventions. Only  $n = 3$  is needed for the disproof.

## 6 Lean formalization

`lean4/Main.lean` (Lean 4.19.0, core library only, no Mathlib) contains:

- *Objects.* Compositions are lists of naturals, and `comps n` lists those of  $n$ . `comps_valid` checks that it lists  $2^{n-1}$  distinct compositions of  $n$  for  $n \leq 8$ . A basis tensor  $S^J \otimes S^K$  is the pair  $(J, K)$ , and a tensor is a formal integer combination of such pairs, compared coefficientwise (`coeffT`). `deltaS I` computes  $\Delta S^I$  as the product of the  $\Delta S_i$  in the tensor-square algebra. An element  $\sum_I x(I) S^I$  of degree  $n$  is given by its coefficient function. `IsPrimitive n x` says that every coefficient of  $\sum_I x(I) (\Delta S^I - S^I \otimes 1 - 1 \otimes S^I)$  vanishes. `PrimDim n m` says that there are  $m$  linearly independent primitives and that every  $m+1$  primitives are linearly dependent, which is the definition of dimension. Coefficients are integers. Linear (in)dependence over  $\mathbb{Z}$  and over  $\mathbb{Q}$  coincide (clear denominators), and rational primitives are rational multiples of integral ones.
- `isPrimitive_three_iff`: Theorem 2’s criterion, for all  $x$ . It uses a support lemma: only finitely many keys can carry nonzero coefficients. The coefficient table on those keys is evaluated by `decide`.
- `primDim_three` : `PrimDim 3 2`. The witnesses are  $\Psi_3$  and  $[S_2, S_1]$ . For the upper bound, any three primitives satisfy an explicit integer relation built from  $2 \times 2$  minors (`dep3`, `three_dep`).
- `primDim_three_unique`: `PrimDim 3 m` implies  $m = 2$ .
- Eulerian numbers: `perms n` enumerates permutations of  $[1, \dots, n]$ . For  $n \leq 5$  it lists  $n!$  distinct permutations, and for  $n = 3$  the membership criterion is checked. `eulerDes0`, `eulerDes1`, `eulerAsc0` and `eulerAsc1` count permutations by descents or ascents, 0- or 1-based. `eulerian_three` gives  $A(3, 1) = 4, 1, 4, 1$ . `eulerian_ne_two` shows that none of the four equals 2 at  $m \leq 6$ .
- `DimClause A`:  $\forall n \geq 1, \text{PrimDim } n \text{ (A } n \text{ 1)}$ . `conjecture_00000000438_false` proves  $\neg \text{DimClause}$  for all four conventions. `conjecture_00000000438_false_degree3` packages  $\dim \text{Prim}(\mathbf{Sym})_3 = 2$  (existence and uniqueness) and  $\dim \neq A(m, 1)$  for every  $m \leq 6$  and every convention.

There is no `sorry`, `native_decide` or added axiom. `#print axioms` reports at most `propext`, `Classical.choice` and `Quot.sound`.

**Not in Lean.** The statement  $A(m, 1) \neq 2$  for  $m \geq 7$  (Lemma 4), the dimensions in degrees  $n \neq 3$ , and the equality of  $\mathbb{Z}$ - and  $\mathbb{Q}$ -dimensions are proved here and checked by `verify.py`. None of them is needed for the refutation at  $n = 3$  under the stated conventions. The undefined “generalized Lie  $n$ -algebra” is not formalized and is not needed.

## 7 Independent verification

`verify.py` uses only the Python 3 standard library. It computes  $\dim \text{Prim}(\mathbf{Sym})_n$  in three ways:

1. as the kernel of  $\bar{\Delta}$ , by exact rational elimination for  $n \leq 8$  and modulo the prime  $2^{61} - 1$  for  $n \leq 10$ ;

2. by duality with QSym, as  $2^{n-1}$  minus the rank of the degree- $n$  quasi-shuffle products  $M_\alpha M_\beta$ , for  $n \leq 7$ ;
3. from the PBW identity above, for  $n \leq 10$ .

All three agree. It computes Eulerian numbers by brute force over all permutations (descents and ascents,  $n \leq 9$ ) and by the closed formula ( $n \leq 10$ ), and it prints the degree-3 computation in full.

## References

- [1] I. M. Gelfand, D. Krob, A. Lascoux, B. Leclerc, V. S. Retakh, J.-Y. Thibon, *Noncommutative symmetric functions*, Adv. Math. 112 (1995), 218–348.
- [2] R. L. Graham, D. E. Knuth, O. Patashnik, *Concrete Mathematics*, 2nd ed., Addison–Wesley, 1994, §6.2.
- [3] OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, A008292 (Eulerian numbers) and A000295 ( $2^n - n - 1$ ).